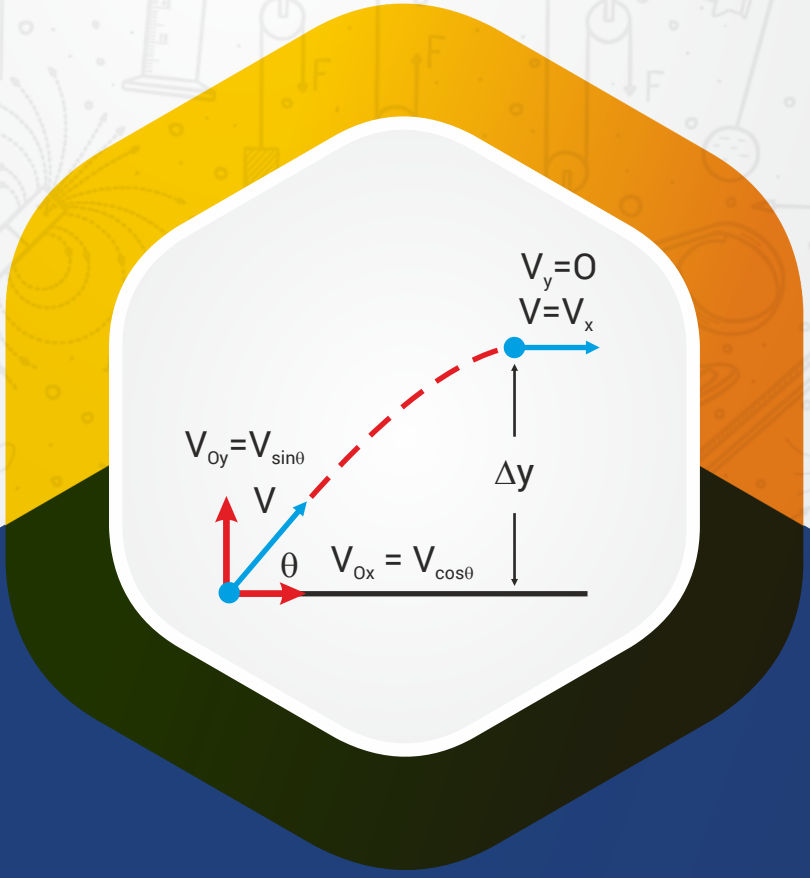
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

PHYSICS

JEE (MAIN+ADVANCED)



Study Material

Kinematics -2D

(English Medium)

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KINEMATICS-2D

KEY CONCEPT

MOTION IN TWO AND THREE DIMENSIONS :

When a particle is moving in space then its motion can be broken up in three co-ordinate axis (x, y& z). The motion in these three directions is governed only by velocity & acceleration in that particular direction and is totally independent of the velocities and acceleration in other directions.

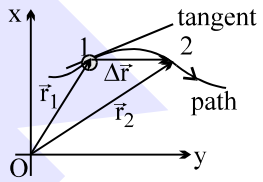
Lets say a particle is moving in space

$\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$

Gives position of particle in space.

VELOCITY

Using the language of calculus, we may write $\vec{v}$ as the derivative



$$\vec{v}=\frac{d\vec{r}}{dt}$$

$$\vec{v}=\frac{d\vec{r}}{dt}=\frac{d}{dt}\left(x\hat{i}+y\hat{j}+z\hat{k}\right)=\left(\frac{dx}{dt}\right)\hat{i}+\left(\frac{dy}{dt}\right)\hat{j}+\left(\frac{dz}{dt}\right)\hat{k}$$

where the scalar components of $\vec{v}$ are

$$v_x=\frac{dx}{dt},\ v_y=\frac{dy}{dt},\ v_z=\frac{dz}{dt}$$

Differentiating $\vec{r}$ w.r.t. time gives us velocity vector of particle at that time.

ACCELRATION

Similarly, if we differentiate $\vec{V}$ w.r.t. time we get acceleration of particle $\vec{a}=\frac{d\vec{V}}{dt}$

$$\vec{a}=\frac{d\vec{v}}{dt}=\frac{dv_x}{dt}\hat{i}+\frac{dv_y}{dt}\hat{j}+\frac{dv_z}{dt}\hat{k}$$

where the scalar components of $\vec{a}$ are

$$a_x=\frac{dv_x}{dt},\ a_y=\frac{dv_y}{dt},\ a_z=\frac{dv_z}{dt}$$

Now, collecting equations of motion relating to x & y axes separately

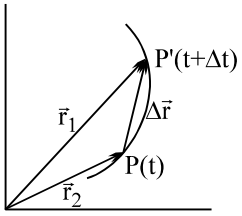
x-axis	y-axis
$V_x=\frac{dx}{dt}$	$V_y=\frac{dy}{dt}$
$a_x=\frac{dV_x}{dt}$	$a_y=\frac{dV_y}{dt}$

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Thus we can see that motion in plane is composed of two straight line motions. **These motions are completely independent of each other**. Only thing connecting them is fact that they are occuring simultaneously.

Velocity is along tangent of path

The direction of the instantaneous velocity $\vec{v}$ of a particle is always tangent to the particle's path at the particle position. $\vec{V} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t}$



The result is the same in three dimensions:

Ex. A particle with velocity $\vec{v}_0 = -2\hat{i} + 4\hat{j}$ (in meters per second) at t = 0 undergoes a constant acceleration $\vec{a}$ of magnitude a = 3 m/s^2 at an angle θ = 127° from the positive direction of the x axis. What is the particle's velocity $\vec{v}$ at t = 5 sec, in unit vector notation ?

Sol. We know that $v = v_0 + at$

now $v_x = v_{0x} + a_x t$ and $v_y = v_{0y} + a_y t$

$a_x = a \cos\theta = (3\ m/s^2)(\cos 127°) = -1.80\ m/s^2$

$a_y = a \sin\theta = (3\ m/s^2)(\sin 127°) = +2.40\ m/s^2$

at time t = 5 sec

$v_x = -2\ m/s + (-1.80\ m/s^2)(5\ sec) = -11\ m/s$

$v_y = 4\ m/s + (2.40\ m/s^2)(5\ sec) = 16\ m/s$

Thus, at t = 5 sec,

$\vec{v} = (-11 m/s)\hat{i} + (16 m/s)\hat{j}$ **Ans.**

Ex. A particle moves in the x–y plane according to the law x = at; y = at(1–αt) where a and α are positive constants and t is time. Find the velocity and acceleration vector. Also find the moment t_0 at which the velocity vector forms angle of 90° with acceleration vector.

Sol. $V_x = a$; $V_y = a - 2a\alpha t \Rightarrow \vec{V} = a\hat{i} + (a - 2a\alpha t)\hat{j}$

$a_x = 0$; $a_y = -2a\alpha \Rightarrow \vec{a} = -2a\alpha\hat{j}$

for 90°, $\vec{V}\cdot\vec{a} = 0$

$-2a\alpha(a - 2a\alpha t_0) = 0$

$1 - 2\alpha t_0 = 0 \Rightarrow t_0 = 1/(2\alpha)$ sec.

PROJECTILE MOTION

We next consider a special case of two-dimensional motion: A particle moves in a vertical plane with some initial velocity $\vec{v}_0$ but its acceleration is always the freefall acceleration $\vec{g}$, which is downward. Such a particle is called a projectile (meaning that it is projected or launched) and its motion is called **projectile motion**.

Assumptions:–

Particle remains close to earth's surface, so acceleration due to gravity remains constant.

Air resistance is neglected.

Distance that projectile travels is small so that earth can be treated as plane surface.

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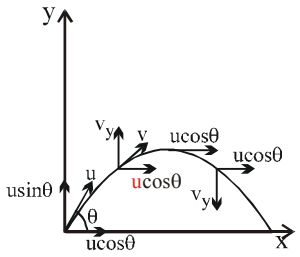
Two straight line motions:–

Our goal here is to analyse projectile motion using the tools for two dimensional motion. This feature allows us to break up a problem involving two dimensional motion into two separate and easier one-dimensional problems,

(a) The horizontal motion is motion with uniform velocity (no effect of gravity)

(b) The vertical motion is motion of uniform acceleration, or freely falling bodies.

Note: In projectile motion, the horizontal motion and the vertical motion are independent of each other, that is either motion does not affects the other.



Treating as two straight line motions:–

The horizontal Motion(x axis):

Because there is no accelration in the horizontal direction, the horizontal component v_x of the projectile velocity remains unchanged from its initial value v_{0x} throughout the motion,

The vertical motion(y axis):

The vertical motion is the motion we discussed for a particle in free fall.

As is illustrated in figure, the vertical component behaves just as for a ball thrown vertically upward. It is directed upward initially and its magnitude steadily decreasing to zero, at the maximum height of the path. The vertical velocity component then reverses direction, and its magnitude becomes larger with time.

x-axis	**y-axis**
Initial velocity(u_x) = ucosθ	Initial velocity(u_y) = usinθ
acceleration(a_x) = 0	acceleration(a_y) = –g
Thus, velocity after time t	Thus, velocity after time t
v_x = ucosθ	v_y = usinθ – gt
Displacement after time t	Displacement after time t
x = ucosθt	y = usinθt – $gt^2/2$

Resultant velocity

$$(\vec{V}_R) = (u\cos\theta)\hat{i} + (u\sin\theta - gt)\hat{j}$$

$$|\vec{V}_R| = \sqrt{u^2\cos^2\theta + (u\sin\theta - gt)^2}$$

& $\tan\alpha = \dfrac{u\sin\theta - gt}{u\cos\theta}$

where α is angle that velocity vector makes with horizontal. Also known as direction or angle of motion

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Time of flight(T)

$$T = \frac{2u\sin\theta}{g}$$

Considering vertical motion

$s_y = 0; u_y = u\sin\theta; a_y = -g$

$$0 = u\sin\theta T - gT^2/2 \Rightarrow T = \frac{2u\sin\theta}{g}$$

Maximum Height(H)

$$H = \frac{u^2\sin^2\theta}{2g}$$

Vertical velocity at maximum height $v_y = 0$

$$0 = u^2\sin^2\theta - 2gH \Rightarrow H = \frac{u^2\sin^2\theta}{2g}$$

Horizontal Range(R)

$$R = \frac{u^2\sin 2\theta}{g} = \frac{2u_x u_y}{g}$$

$$\text{Total time } T = \frac{2u\sin\theta}{g}$$

Velocity in horizontal direction $u_x = u\cos\theta$

Total displacement in horizontal direction $R = u\cos\theta T$

$$R = \frac{u^2\sin 2\theta}{g}$$

Note:– *For complementry angles i.e. $\theta + \alpha = 90°$, the range is same for same projection speed but maximum height and time of flight are different.*

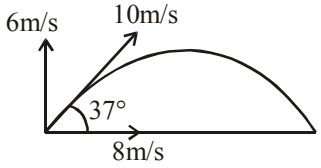
Ex. A body is thrown with initial velocity 10m/sec. at an angle 37° from horizontal. Find (Take : $g = 10m/s^2$)

(i) Time of flight (ii) Maximum height.

(iii) Range (iv) Position vector at t = 1 sec.

Ans. (i) 1.2 sec, (ii) 1.8 m, (iii) 9.6 m, (iv) $(8\hat{i} + \hat{j})$

Sol. (i) Time of flight $T = \frac{2u\sin\theta}{g} = \frac{2\times 10\times\frac{3}{5}}{10} = \frac{6}{5} = 1.2 \text{ sec}$



(ii) Maximum height $H = \frac{u^2\sin^2\theta}{2g} = \frac{100\left(\frac{9}{25}\right)}{2\times 10} = \frac{9}{5} = 1.8 \text{ m}$

(iii) $R = \frac{u^2\sin 2\theta}{g} = \frac{100\times 2\times\frac{3}{5}\times\frac{4}{5}}{10} = \frac{240}{25} = 9.6\text{m}$

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(iv) $x = 8 \times 1 = 8$ m

$$y = 6 \times 1 - \frac{1}{2} \times 10 \times (1)^2 = 1\text{m}$$

$$\vec{r} = 8\hat{i} + \hat{j}$$

Maximum Range

$$R = \frac{u^2 \sin 2\theta}{g}$$

for $\theta = 45°$, R is maximum

$$R_{max} = \frac{u^2}{g}$$

Ex. A person can throw a ball vertically upto maximum height of 20 mt. How far can he throw the ball.

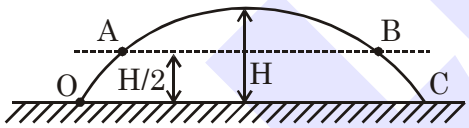
Sol. $H_{max} = \frac{u^2}{2g}$

$\therefore u = 20$ m/s

$$R_{max} = \frac{u^2}{g} = 40 \text{ m}$$

Ex. A particle is projected with a speed u at an angle θ with horizontal. Find the average velocity of projectile for the period during which it crosses half of maximum height.

Sol.



avg. velocity is a vector

Let t_1 is time taken by particle to travel from A to B

First we will find vertical component.

For motion along AB

$$(\vec{V}_y)_{avg} = \frac{\text{Total Displacement along y direction}}{\text{Total time}} = \frac{0}{t_1} = 0$$

For Horizontal component

$$(\vec{V}_x)_{avg} = \frac{\text{Total Displacement along x direction}}{\text{Total time}}$$

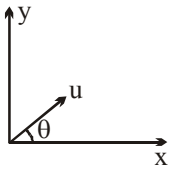
$$(\vec{V}_x)_{avg} = \frac{(u\cos\theta)t_1}{t_1} = u\cos\theta$$

So average velocity for motion along AB is ucosθ

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EQUATION OF TRAJECTORY

Lets say point of projection is our origin and horizontal direction is x-axis and vertically upwards is positive y-axis.



We know $x = u\cos\theta t$

$$\therefore \quad t = \frac{x}{u\cos\theta} \qquad .....(1)$$

$$\text{also } y = u\sin\theta t - \frac{1}{2}gt^2 \qquad .....(2)$$

Putting value of 't' from eq. (1) in eq. (2), we get

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$$

Ex. A particle is projected with a velocity 10 m/s at an angle 37° to the horizontal. Find the location at which the particle is at a height 1m from point of projection. ($g = 10\ m/s^2$)

Ans. 1.6 m, 8 m.

Sol. $$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$$

for $y = 1; \theta = 37°; u = 10$ m/s

$$1 = \frac{3}{4}x - \frac{10x^2}{2\times100\times\left(\frac{16}{25}\right)}$$

$$1 = \frac{3}{4}x - \frac{5}{64}x^2$$

$5x^2 - 48x + 64 = 0$

$5x^2 - 40x - 8x + 64 = 0$

$x = 8m, 1.6m$

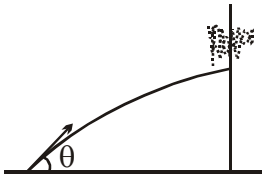
Ex. We have a hose pipe which disposes water at the speed of 10 ms^{-1}. The safe distance from a building on fire, on ground is 5 m. How high can this water go? (take : $g = 10\ ms^{-2}$)

Sol. Here we must understand that taking range of projectile as 10m and making projectile hit the building when it is at maximum height is wrong. By doing this we are not acheiving maximum y for given x = 5m. This just makes highest point of path to like on x = 5, But there may be other path for which y will be maximum for given x. This problem will be solved by using equation of trajectory by putting x = 5m and maximising y by varying θ.

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$$

Putting $x = 5m$ we get

$$y = 5\tan\theta - \frac{10\times25\sec^2\theta}{2\times100}$$



$5\tan^2\theta - 20\tan\theta + (4y + 5) = 0$

for real roots discriminant must be positive.

$400 - 4 \times 5\,(4y + 5) > 0$

Solving $3.75 \geq y$

hence maximum $y = 3.75$ m

If we have taken range as 10 m then angle of projection will be $\theta = 45°$ corsponding maximum hight H = 2.5m which is smaller than our answer.

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Ex. A projectile is fired horizontaly with a velocity of 98 m/s from the top of a hill 490 m high. Find
(i) Time to reach ground
(ii) The horizontal distance from foot of hill to ground.
(iii) The speed with which it hits the ground.

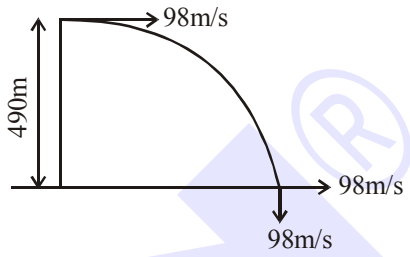
Ans. (i) 10 sec, (ii) 980 m, (iii) $98\sqrt{2}$ m/se

Sol. $u_y = 0; u_x = 98$ m/s
on y-axis

$$s = ut + \frac{1}{2}at^2$$

$$-490 = 0 - \frac{1}{2} \times 9.8 \times t^2$$

$$t = \sqrt{\frac{490 \times 2}{9.8}} = 10 \text{ sec}$$



on x-axis for horizontal displacement,

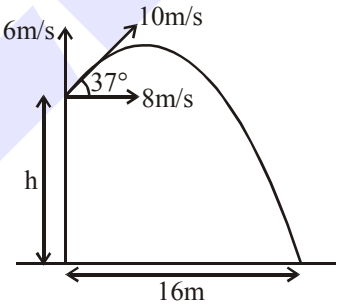
$$R = u_x t = 98 \times 10 = 980 \text{ m}$$

$v_x = 98$ m/s
$v_y = 0 - 9.8 \times 10 = -98$ m/s
So speed = $98\sqrt{2}$ m/s

Ex. A ball is thrown from the top of a tower with an initial velocity of 10m/s at an angle 37° above the horizontal, hits the ground at a distance 16m from the base of tower. Calculate height of tower. [$g = 10$ m/s^2]

Ans. 8 m

Sol. Time to reach ground is $= \frac{16}{8} = 2$ sec



on y-axis (for height of tower)

$$-h = 6 \times 2 - \frac{1}{2} \times 10 \times (2)^2$$

$-h = 12 - 22$
$h = 8$ m

Ex. Prithvi missile is fired to destroy an enemy military base situated on same horizontal level, situated 99 km away. The missile rises vertically for 1 km & then for remainder of flight, it follows parbolic path like a free body under earth's gravity, at an angle of 45°. Calculate its velocity at beginning of parabolic path. ($g = 10$ ms^{-2})

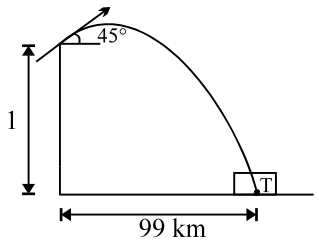
Sol. for horizontal motion time t

$$t = \frac{99 \times 10^3}{u \cos 45°}$$

for vertical

$$-1 \times 10^3 = u \sin 45° \, t - \frac{1}{2} \times 10 \times t^2$$

$$1 \times 10^3 + \frac{u \sin 45°}{u \sin 45°} \times 99 \times 10^3 = \frac{10}{2} \times \frac{(99 \times 10^3)^2 \times 2}{u^2}$$



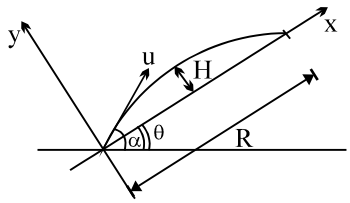
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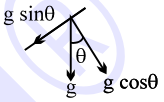
$$u^2 = \frac{(99\times10^3)^2\times10}{100\times10^3}$$

$$u = 99\times10^3\sqrt{\frac{1}{10^4}} = 990\ ms^{-1}$$

PROJECTION ON INCLINED PLANE

There is an inclined plane making an angle θ with horizontal. A particle is projected at an angle α from horizontal.





x-axis

$u_x = u\cos(\alpha-\theta)$

$a_x = -g\sin\theta$

vel. at any time t

$v_x = u\cos(\alpha-\theta) - g\sin\theta\, t$

y-axis

$u_y = u\sin(\alpha-\theta)$

$a_y = -g\cos\theta$

vel. at any time t

$v_y = u\sin(\alpha-\theta) - g\cos\theta\, t$

Time of flight

Displacement in y direction $s_y = 0$

$$0 = u\sin(\alpha-\theta)\,T - \frac{1}{2}g\cos\theta\,T^2$$

$$T = \frac{2u\sin(\alpha-\theta)}{g\cos\theta}$$

Maximum distance of particle from inclined plane

Point where $v_y = 0$ is max.-distance from inlcine plane

$(0)^2 = u^2\sin^2(\alpha-\theta) - 2g\cos\theta\, H$

$$H = \frac{u^2\sin^2(\alpha-\theta)}{2g\cos\theta}$$

Range along the inclined plane

$$s_x = u_xT + \frac{1}{2}a_xT^2$$

$$R = \frac{u\cos(\alpha-\theta)\times2u\sin(\alpha-\theta)}{g\cos\theta} - \frac{g\sin\theta\times2\times2u^2\sin^2(\alpha-\theta)}{2g^2\cos^2\theta}$$

$$= \frac{2u^2\sin(\alpha-\theta)[\cos(\alpha-\theta)\cos\theta - \sin\theta\sin(\alpha-\theta)]}{g\cos^2\theta}$$

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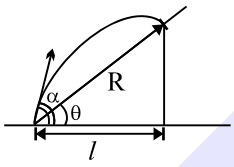
$$R = \frac{2u^2 \sin(\alpha-\theta)\cos\alpha}{g\cos^2\theta}$$

$$R = \frac{u^2[\sin(2\alpha-\theta)-\sin\theta]}{g\cos^2\theta}$$

Alternate Method

$l = u\cos\alpha\, T$

$$R = \frac{l}{\cos\theta}$$



$$R = \frac{u\cos\alpha}{\cos\theta} \times \frac{2u\sin(\alpha-\theta)}{g\cos\theta}$$

$$R = \frac{2u^2\sin(\alpha-\theta)\cos\alpha}{g\cos^2\theta}$$

Note : Presence of incline plane does not affect the path of projectile in any way.

Maximum Range :

$$R = \frac{u^2[\sin(2\alpha-\theta)-\sin\theta]}{g\cos^2\theta}$$

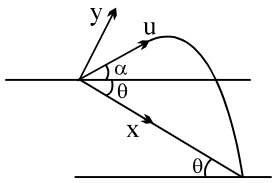
For max. range $2\alpha-\theta = \frac{\pi}{2} \Rightarrow \alpha = \frac{\pi}{4}+\frac{\theta}{2}$

so $R_{max} = \frac{u^2}{g(1+\sin\theta)}$

Projection from top of incline plane:

Incline plane is at an angle θ with horizontal and a particle is projected at an angle α from horizontal.

In all formulae replace θ with –θ



$$H = \frac{u^2\sin^2(\alpha+\theta)}{2g\cos\theta}$$

$$T = \frac{2u\sin(\alpha+\theta)}{g\cos\theta}$$

$$R = \frac{2u^2\sin(\alpha+\theta)\cos\alpha}{g\cos^2\theta}$$

$$R_{max} = \frac{u^2}{g(1-\sin\theta)} \text{ and } \alpha = \frac{\pi}{4}-\frac{\theta}{2}$$

Note : If a particle strikes the incline plane ⊥ then its comp. of velocity along incline must be zero.

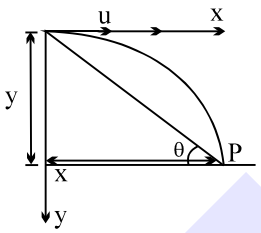
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Ex. A particle is projected horizontally with a speed u from the top of a plane inclined at an angle θ with the horizontal. How far along the plane, from the point of projection will particle strike the plane?

Sol.

x-axis	**y-axis**
$u_x = u$	$u_y = 0$
$a_x = 0$	$a_y = g$
$x = ut$	$y = \frac{gt^2}{2}$

$$\Rightarrow \quad y = \frac{g x^2}{2u^2}$$



$$\text{also } \frac{y}{x} = \tan\theta \Rightarrow x\tan\theta = \frac{g x^2}{2u^2}$$

$$x = 0, \ \frac{2u^2\tan\theta}{g}$$

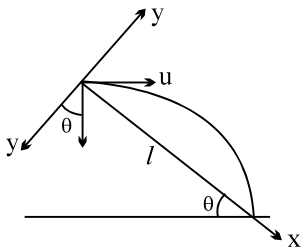
$$x = \frac{2u^2\tan\theta}{g}$$

$$\Rightarrow y = \frac{2u^2\tan^2\theta}{g}$$

$$\text{dist. } l = \sqrt{x^2 + y^2}$$

$$l = \frac{2u^2\tan\theta\sec\theta}{g}$$

Alternate Method.

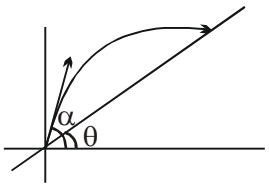


$$R = \frac{2u^2\sin(\alpha+\theta)\cos\alpha}{g\cos^2\theta} \quad \text{as } \alpha = 0$$

$$R = 2u^2\tan\theta\sec\theta$$

Ex. A particle is projected up an inclined plane. Plane is inclined at an angle θ with horizontal and particle is projected at an angle α with horizontal. If particle strikes the plane horizontally prove that $\tan\alpha = 2\tan\theta$

Sol. We know time of flight



$$T = \frac{2u\sin(\alpha+\theta)}{g\cos\theta}$$

considering vertical motion

$u = v\sin\alpha$

$a = -g$

$v = 0$

$$\therefore \ T = \frac{u\sin\alpha}{g} = \frac{2u\sin(\alpha-\theta)}{g\cos\theta}$$

$\sin\alpha\cos\theta = 2\sin\alpha\cos\theta - 2\cos\alpha\sin\theta$

$2\cos\alpha\sin\theta = \sin\alpha\cos\theta$

$2\tan\theta = \tan\alpha$

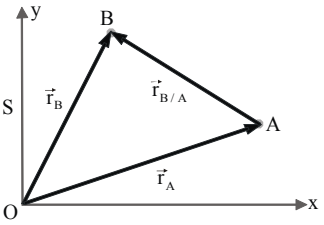
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Relative Motion

Motion of a body can only be observed, when it changes its position with respect to some other body. In this sense, motion is a relative concept. To analyze motion of a body say A, therefore we have to fix our reference frame to some other body say B. The result obtained is motion of body A relative to body B.

Relative position, Relative Velocity and Relative Acceleration



Let two bodies represented by particles A and B at positions defined by position vectors $\vec{r}_A$ and $\vec{r}_B$, moving with velocities $\vec{v}_A$ and $\vec{v}_B$ and accelerations $\vec{a}_A$ and $\vec{a}_B$ with respect to a reference frame S. For analyzing motion of terrestrial bodies the reference frame S is fixed with the ground.

The vectors $\vec{r}_{B/A}$ denotes position vector of B relative to A. Following triangle law of vector addition, we have

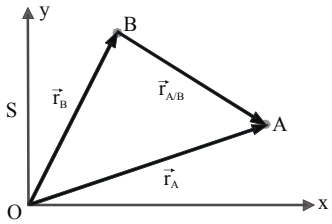
$$\vec{r}_B = \vec{r}_A + \vec{r}_{B/A} \qquad ...(i)$$

First derivatives of $\vec{r}_A$ and $\vec{r}_B$ with respect to time equals to velocity of particle A and velocity of particle B relative to frame S and first derivative of $\vec{r}_{B/A}$ with respect to time defines velocity of B relative to A.

$$\vec{v}_B = \vec{v}_A + \vec{v}_{B/A} \qquad ...(ii)$$

Second derivatives of $\vec{r}_A$ and $\vec{r}_B$ with respect to time equals to acceleration of particle A and acceleration of particle B relative to frame S and second derivative of $\vec{r}_{B/A}$ with respect to time defines acceleration of B relative to A.

$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A} \qquad ...(iii)$$



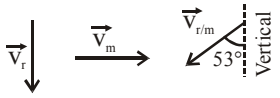
In similar fashion motion of particle A relative to particle B can be analyzed with the help of adjoining figure. You can observe in the figure that position vector of A relative to B is directed from B to A and therefore

$$\vec{r}_{B/A} = -\vec{r}_{A/B},\ \vec{v}_{B/A} = -\vec{v}_{A/B} \text{ and } \vec{a}_{B/A} = -\vec{a}_{A/B}.$$

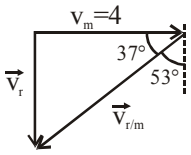
The above equations elucidate that how a body A appears moving to another body B is opposite to how body B appears moving to body A.

Ex. A man when standstill observes the rain falling vertically and when he walks at 4 km/h he has to hold his umbrella at an angle of 53° from the vertical. Find velocity of the raindrops.

Sol. Assigning usual symbols $\vec{v}_m$, $\vec{v}_r$ and $\vec{v}_{r/m}$ to velocity of man, velocity of rain and velocity of rain relative to man, we can express their relationship by the following eq. $\vec{v}_r = \vec{v}_m + \vec{v}_{r/m}$



The above equation suggests that a standstill man observes velocity $\vec{v}_r$ of rain relative to the ground and while he is moving with velocity $\vec{v}_m$, he observes velocity of rain relative to himself $\vec{v}_{r/m}$. It is a common intuitive fact that umbrella must be held against $\vec{v}_{r/m}$ for optimum protection from rain. According to these facts, directions of the velocity vectors are shown in the adjoining figure.



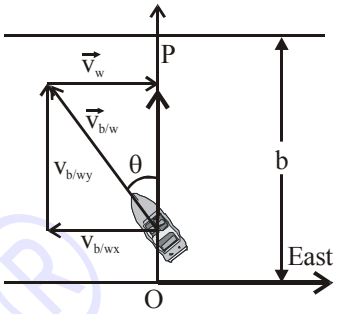
The addition of velocity vectors is represented according to the above equation. From the figure we have

$v_r = v_m \tan 37° = 3$ km/h **Ans.**

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Ex. A boat can be rowed at 5 m/s on still water. It is used to cross a 200 m wide river from south bank to the north bank. The river current has uniform velocity of 3 m/s due east.
(a) In which direction must it be steered to cross the river perpendicular to current?
(b) How long will it take to cross the river in a direction perpendicular to the river flow?
(c) In which direction must the boat be steered to cross the river in minimum time? How far will it drift?

Sol. (a) Velocity of a boat on still water is its capacity to move on water surface and equals to its velocity relative to water.

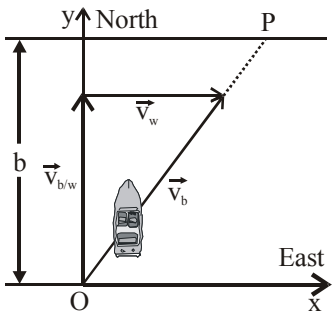


$\vec{v}_{b/w}$ = Velocity of boat relative to water = Velocity of boat on still water

On flowing water, the water carries the boat along with it. Thus velocity $\vec{v}_b$ of the boat relative to the ground equals to vector sum of $\vec{v}_{b/w}$ and $\vec{v}_w$. The boat crosses the river with the velocity $\vec{v}_b$.

$$\vec{v}_b = \vec{v}_{b/w} + \vec{v}_w$$

(b) To cross the river perpendicular to current the boat must be steered in a direction so that one of the components of its velocity ($\vec{v}_{b/w}$) relative to water becomes equal and opposite to water flow velocity $\vec{v}_w$ to neutralize its effect. It is possible only when velocity of boat relative to water is grater than water flow velocity. In the adjoining figure it is shown that the boat starts from the point O and moves along the line OP (y-axis) due north relative to ground with velocity $\vec{v}_b$. To achieve this it is steered at an angle θ with the y-axis.



$v_{b/w} \sin\theta = v_w \rightarrow 5\sin\theta = 3 \Rightarrow \theta = 37°$ **Ans.**

(c) The boat will cover river width b with velocity

$v_b = v_{b/wy} = v_{b/w} \sin 37° = 4$ m/s in time t, which is given by

$t = b / v_b \rightarrow t = 50s$ **Ans.**

(d) To cross the river in minimum time, the component perpendicular to current of its velocity relative to ground must be kept to maximum value. It is achieved by steering the boat always perpendicular to current as shown in the adjoining figure. The boat starts from O at the south bank and reaches point P on the north bank. Time t taken by the boat is given by

$t = b / v_{b/w} \rightarrow t = 40s$ **Ans.**

Drift is the displacement along the river current measured from the starting point. Thus, it is given by the following equation. We denote it by x_d.

$$x_d = v_{bx} t$$

Substituting $v_{bx} = v_w = 3$ m/s, from the figure, we have

$x_d = 120$ m **Ans.**

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